

see what's up, to transact (yup, that's right: e-commerce right on the 'fourth screen') and to make intelligent energy resource decisions," Bell writes in an article.

There are signs of these dreams materialising soon. For one, the industry is slowly converging on the protocols and standards for smart appliances and home networking. Zigbee, for example, is emerging as a popular standard on this front. Newer applications and smart appliances are also emerging on the scene. GE's product line is one typical example.

"Home energy management (HEM), enabled by the smart grid, is an extension of smart meter deployments that provide interesting benefits for residential customers and utilities. IDC found that customers reduce overall energy use by 4 to 15 per cent when they receive real-time feedback on power consumption. For utilities, HEM solutions open new opportunities for strengthening customer relationships, managing power loads, defending against new competitors and realising new revenue from value-added services," says Raina of Cisco, which has launched some solutions in this space.

Cisco's HEM gives residential consumers the ability to see and understand their energy use—and its costs—in real time. The consumer interface is the Home Energy Controller (HEC), a countertop display with an LCD touchscreen that can communicate with other home devices—such as a smart meter, smart plugs and programmable thermostats—to help optimise in-home energy management. The HEC has the potential to do for residential consumers what hybrid drivers can do each time they sit behind the wheel.

At this year's *Consumer Electronics Show*, Intel revealed its

intelligent in-home device, described as the 'fourth screen' for the home (the first three being television, computers and mobile phones). The device is powered by an Intel Atom processor and provides a centralised dashboard manifested on a stylish, technologically-advanced OLED touch screen. It helps the consumers understand their power consumption patterns and automatically schedule and control devices using smart appliances, programmable thermostats, etc.

Industry majors like Microsoft and Google also offer power management solutions. Rumours are that Apple will also be entering this space soon, as it has filed two related patent applications. The proposed energy management system would allow consumers to reduce electricity bills by controlling and maximising how power is allocated to home electronics such as computers, cell phones and even iPods.

Apparently, the home energy management dashboard is not based on any new technology. Data would be transferred through existing home or business wiring with the HomePlug Powerline Network—a plug-in device that essentially turns outlets into high-speed Internet sources. The system would give consumers real-time information on their energy use, how much it costs and the opportunity to do something about it. Whether all this would require a new device or an application that can run on existing ones is not clear yet.

Such innovative devices and solutions that make it fun, exciting and easy for customers to understand and manage their energy consumption would take demand-side management to the next level. To begin with, we could start calling the initiative as 'smart homes' rather than in staid technical terms!

Smarter Smart Grids: Looking Into The Future

Current technological developments, research and discussions happening world over indicate that smart grids are going to become smarter than ever before

Homes that give back. One of the key features of a smart grid is its ability to combine power produced from distributed power sources. Hence, theoretically, every home can be producer. If the house is able to produce more energy than it consumes, through solar and other renewable power sources, it can give back to the grid, and get discounts on its bill too. Such an idea was also presented by GE when it demonstrated its vision of a zero net energy house.

Grids that crystal-gaze. Instead of merely reacting fast to extraordinary situations, grids of the future will even be capable of anticipating peaks or troughs in demand and planning the distribution accordingly.

Smart grid 2.0. Echelon's CEO, Robert A. Dolin believes

that smart grid technology is more than mere two-way communications and remote meter reading. The Smart Grid 2.0 puts intelligence and communications in devices throughout the grid from distribution equipment (meters, transformers, capacitor banks, etc) to commercial electrical devices (chillers, boilers, air handlers, lighting, etc) and home appliances like electric water heaters, air-conditioners, and rooftop solar arrays. These smart devices can now become a part of the grid, able to interact in real time to changing conditions on the grid.

The Smart Grid 2.0 uses the grid network and the devices connected to it as a communicating, intelligent system for the delivery of additional services and increased operational efficiency, such as demand response programs. Another sig-